

Lecture 9 - Difference in Differences

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- Difference-in-Differences Framework
- The most famous example: Card and Krueger (1994) Minimum Wage in NJ vs. PA
- Violations of parallel trends:
 - Ashenfelter Dip
- Event study:
 - Minimum wage changes in different states, longer panels
 - mover design in estimating firm wage premiums

Motivating Example

Difference in Differences/Event Studies - CK 1994 Example

- We were interested in studying the effect of the minimum wage on employment of fast food restaurants. We had in mind a model like:

$$y_{rs1} = \beta_2 \text{Min}_{s1} + \delta_1 + \alpha_{rs} + \epsilon_{rs1}$$

$$y_{rs2} = \beta_2 \text{Min}_{s2} + \delta_2 + \alpha_{rs} + \epsilon_{rs2}$$

where y_{rst} is employment of restaurant r in state s in period $t = 1, 2$, Min_{st} is the minimum wage in state s in period t , δ_1 and δ_2 are constants that reflect the general state of the economy in periods 1 and 2, α_{rs} is a restaurant specific effect, capturing things like location of the outlet, and ϵ_{rst} is a transitory error component.

- Note the difference given any restaurant r in state s :

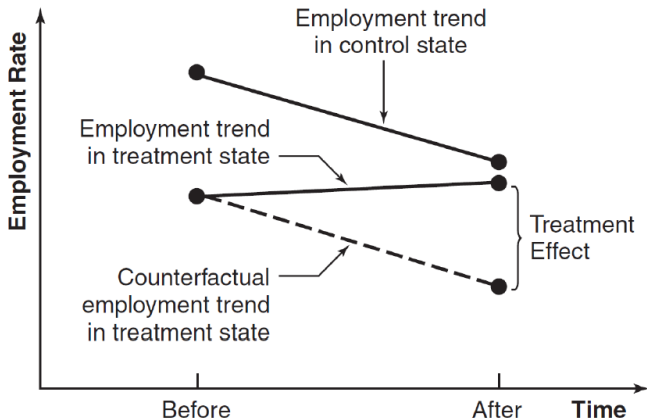
$$y_{rs2} - y_{rs1} = \beta_2 \Delta \text{Min}_s + (\delta_2 - \delta_1) + \Delta \epsilon_{rs}$$

To get a consistent estimate of β_2 , we need:

- 1 $\Delta \epsilon_{rs} \perp (\delta_2 - \delta_1), \Delta \text{Min}_s$, satisfied if ϵ_{rst} are exogenous transitory errors.
- 2 A way to estimate the trend $(\delta_2 - \delta_1)$.

Motivating Example

- The idea of the difference-in-differences model is to select a control group without the policy change, and have the same $(\delta_2 - \delta_1)$ – in the absence of the policy intervention, the treated and the control would have exhibited the same trend.



Framework: Difference of Differences

- Consider the causal model:

$$y_{it} = x'_{it}\beta + \nu_{it}$$

The assumption in the basic panel data approach is that the “problem” with ν_{it} is confined to the time-invariant component – α_i .

- We decompose $\nu_{it} = \alpha_i + \epsilon_{it}$ and **assume** $E[x_{it}\epsilon_{is}] = 0$ (strict exogeneity). It can be very hard to evaluate this assumption.
- There is a case where it is easier. Suppose the x_{it} we're interested in has a specific “design feature”:

$$\begin{aligned}x_{it} &= 0 & i \in \text{group } 0, \text{ all } t \\x_{it} &= 0 & i \in \text{group } 1, t < t^* \\x_{it} &= 1 & i \in \text{group } 1, t \geq t^*\end{aligned}$$

Framework: Difference of Differences

- In this case, x_{it} is a dummy variable that is 0 for all members of group 0, is also 0 for all members of group 1 in the periods up to $t^* - 1$, and then **“turns on” in period t^* for group 1**:

$$x_{it} = D_i \times 1[t \geq t^*]$$

In this design we call group 0 the “comparison”/control group and group 1 the “treatment” group.

- We are also concerned that mean outcomes change over time. Define a set of time dummies $\{T_{1it}, T_{2it} \dots\}$, (i.e., T_{sit} is an indicator for period s).

$$T_{sit} = \begin{cases} 1 & s = t \\ 0 & \text{otherwise} \end{cases}$$

in fact you can simplify the time dummy as T_{st} . The subscript (i, t) serves to highlight the unit of observation in this panel dataset is a person $\times$ year observation.

Framework: Difference of Differences

- So we can write our causal model as:

$$\text{Among } D_i = 0 : y_{it} = T_{1it}\delta_1 + \dots + T_{Sit}\delta_S + \nu_{it} \quad (1)$$

$$\begin{aligned} \text{Among } D_i = 1 : y_{it} = & T_{1it}\delta_1 + \dots + T_{Sit}\delta_S \\ & + \underbrace{1[t \geq t^*]} \times \theta + \nu_{it} \end{aligned} \quad (2)$$

$$\begin{aligned} \text{Pooled : } y_{it} = & T_{1it}\delta_1 + \dots + T_{Sit}\delta_S + \\ & + \underbrace{D_i \times 1[t \geq t^*]} \times \theta + \nu_{it} \end{aligned} \quad (3)$$

$$\text{simplified notation } y_{it} = \delta_t + D_i 1[t \geq t^*] \times \theta + \nu_{it} \quad (4)$$

In the simplified notation (also called “effect notation”) the δ_t variables mean a combination of a dummy for period t and a separate coefficient for that dummy!

- Note the treatment and the control may have different trends – diff $\{\delta_t\}$ in (1) and (2). How we write the models assume **parallel trend** – $\{\delta_t\}$ in model (1) among the control is the same $\{\delta_t\}$ in model (2) among the treated.

Diff-in-Diff: Assumptions

Causal model:

$$\begin{aligned}y_{it} &= T_{1it}\delta_1 + \dots + T_{Sit}\delta_S + D_i 1[t \geq t^*]\theta + \nu_{it} \\ &= \delta_t + D_i 1[t \geq t^*]\theta + \nu_{it} \text{ (simplified notation)}\end{aligned}$$

Basic idea:

- 1 Assume that $\nu_{it} = \alpha_i + \epsilon_{it}$ (permanent & transitory errors)
- 2 Assume “selection” is based on α_i , not the transitory ϵ'_{it} s. Then we can use the pre-treatment periods to estimate the relationship between D_i and α_i , then use that to “back out” the bias in the post-treatment periods.

(a) Diff between the treated and control post treatment

$$y_{it} = \delta_t + D_i 1[t \geq t^*] \theta + \alpha_i + \epsilon_{it}$$

WLOG assume $t = t^* = 2$ is a post period.

$$y_{i2} = \delta_2 + D_i \theta + \alpha_i + \epsilon_{i2}$$

What do you get if running a cross-sectional regression of y_{i2} on $(1, D_i)$ at $t = 2$? We would estimate

$$\theta^* = \theta + \pi,$$

where bias π comes from the auxiliary regression:

$$\begin{aligned} \alpha_i &= \pi_0 + D_i \pi + \xi_i \\ \Rightarrow \pi &= E[\alpha_i | D_i = 1] - E[\alpha_i | D_i = 0] \end{aligned}$$

(a) Diff between the treated and control post treatment

Problem: $E[\alpha_i|D_i = 1] \neq E[\alpha_i|D_i = 0]$ (non-random selection)

- The bias term $\pi = E[\alpha_i|D_i = 1] - E[\alpha_i|D_i = 0]$ represents the “selection bias”.
- Comparing the treatment and control groups in a single period is not sufficient. The idea of diff-in-diff is to compute the change in the difference between the two groups, from pre to post treatment.

(b) Change in y among the treated?

What if we use only the change in y for the program group?

$$y_{i2} - y_{i1} = \underbrace{\delta_2 - \delta_1}_{\text{change in } \delta} + \underbrace{\theta}_{\text{treatment effect}} + \epsilon_{i2} - \epsilon_{i1}$$

Problem: can't distinguish change in δ over time from the “treatment effect” θ .

Diff between the treated and control in pre vs. post

$$y_{it} = \delta_t + D_i 1[t \geq t^*] \theta + \alpha_i + \epsilon_{it}$$

- Let's consider the mean difference between program and comparison groups in any period:
- Any pre-period ($s < t^*$):

$$\Delta(s) = E[y_{is} | D_i = 1] - E[y_{is} | D_i = 0] = \pi$$

- Any post-period ($\tau \geq t^*$):

$$\Delta(\tau) = E[y_{i\tau} | D_i = 1] - E[y_{i\tau} | D_i = 0] = \pi + \theta$$

$\Rightarrow$ can estimate $\theta = \Delta(\tau) - \Delta(s)$ - the ***“difference in differences”***

Diff between the treated and control in pre vs. post

- How can we do that with a pooled regression, given data at a pre-period s and a post-period τ ?
- Plug in $\alpha = \pi_0 + D_i\pi + \epsilon_{it}$ to the causal model
 $y_{it} = \delta_t + D_i 1[t \geq t^*]\theta + \alpha_i + \epsilon_{it}$

$$y_{it} = T_{sit}\delta_s + T_{\tau it}\delta_\tau + D_i\pi + D_i \times 1[t \geq t^*]\theta + \epsilon_{it} \quad (5)$$

- the constant in the auxiliary regression: $\pi_0 = E[\alpha|D=0]$ is added to both (δ_s, δ_τ) in the regression above w/o a constant.
- One could also estimate (5) with a constant, but drop the dummy for the pre-period. The coef on post-period would be $\delta_\tau - \delta_s$.

We know the regression will fit the means for all groups that are separately identified.

Diff between the treated and control in pre vs. post

$$y_{it} = \underbrace{T_{sit}\delta_s}_{\text{pre}} + \underbrace{T_{\tau it}\delta_\tau}_{\text{post}} + \underbrace{D_i\pi}_{\text{select}} + \underbrace{D_i1[t \geq t^*]\theta}_{\text{diff-in-diff}} + v_{it}$$

There are 4 dummies for 4 groups. So the 4 groups will be fit perfectly!

Given s : pre, and t : post,

$$E[y_{is}|D_i = 0] = \delta_s$$

$$E[y_{is}|D_i = 1] = \delta_s + \pi$$

$$E[y_{i\tau}|D_i = 0] = \delta_\tau$$

$$E[y_{i\tau}|D_i = 1] = \delta_\tau + \pi + \theta$$

$$\begin{aligned}\Rightarrow \Delta(\tau) &= E[y_{i\tau}|D_i = 1] - E[y_{i\tau}|D_i = 0] \\ &= \pi + \theta\end{aligned}$$

$$\begin{aligned}\Delta(s) &= E[y_{is}|D_i = 1] - E[y_{is}|D_i = 0] \\ &= \pi\end{aligned}$$

$$\theta = \Delta(\tau) - \Delta(s)$$

Card and Krueger (1994)

- Card-Krueger study of the increase in the minimum wage in New Jersey.
 - NJ min. wage rose from \$4.25/hr to \$5.05/hr on April 1, 1992
 - surveys of restaurants in NJ and PA before and after
- Unit of observation: $(r, s, t) = (\text{restaurant}, \text{state}, \text{time})$
 - r : KFC, Burger King...
 - s : NJ vs. PA
 - t : early 1992 (wave1 before min. wage increase), late 1992 (wave2 after min. wage increase)
- Is PA a good control group for NJ?

TABLE 2—MEANS OF KEY VARIABLES

Variable	Stores in:		<i>t</i> ^a
	NJ	PA	
1. <i>Distribution of Store Types (percentages):</i>			
a. Burger King	41.1	44.3	-0.5
b. KFC	20.5	15.2	1.2
c. Roy Rogers	24.8	21.5	0.6
d. Wendy's	13.6	19.0	-1.1
e. Company-owned	34.1	35.4	-0.2
2. <i>Means in Wave 1:</i>			
a. FTE employment	20.4 (0.51)	23.3 (1.35)	-2.0
b. Percentage full-time employees	32.8 (1.3)	35.0 (2.7)	-0.7
c. Starting wage	4.61 (0.02)	4.63 (0.04)	-0.4
d. Wage = \$4.25 (percentage)	30.5 (2.5)	32.9 (5.3)	-0.4
e. Price of full meal	3.35 (0.04)	3.04 (0.07)	4.0
f. Hours open (weekday)	14.4 (0.2)	14.5 (0.3)	-0.3
g. Recruiting bonus	23.6 (2.3)	29.1 (5.1)	-1.0

Card and Krueger (1994)

3. Means in Wave 2:

a. FTE employment	21.0 (0.52)	21.2 (0.94)	-0.2
b. Percentage full-time employees	35.9 (1.4)	30.4 (2.8)	1.8
c. Starting wage	5.08 (0.01)	4.62 (0.04)	10.8
d. Wage = \$4.25 (percentage)	0.0	25.3 (4.9)	—
e. Wage = \$5.05 (percentage)	85.2 (2.0)	1.3 (1.3)	36.1
f. Price of full meal	3.41 (0.04)	3.03 (0.07)	5.0
g. Hours open (weekday)	14.4 (0.2)	14.7 (0.3)	-0.8
h. Recruiting bonus	20.3 (2.3)	23.4 (4.9)	-0.6

Notes: See text for definitions. Standard errors are given in parentheses.

^aTest of equality of means in New Jersey and Pennsylvania.

- Assumed causal model is:

$$\begin{aligned}y_{rst} &= T_{1t}\delta_1 + T_{2t}\delta_2 + NJ_{rs}Post_t\beta_1 + v_{rst} \\&= \delta_1 + T_{2t}(\delta_2 - \delta_1) + NJ_{rs}Post_t\beta_1 + v_{rst} \\&= \delta_1 + Post_t\delta_2^0 + NJ_{rs}Post_t\beta_1 + v_{rst}\end{aligned}$$

where the unit of observation is $(r, s, t) = (\text{restaurant, state, time})$. Think of the " i " = (r, s) Burger King could have locations in both NJ and PA.

- y_{rst} is employment at restaurant r in state s in period $t = 1, 2$,
- $NJ_{rs} = 1$ if the restaurant (r, s) is in state NJ.
- $Post_t = 1$ if the time $t \geq t^*$ the year in which minimum wage was increased in NJ.
- $NJ_{rs}Post_t$ is a dummy for the "high" min. wage in effect. That is,

$$NJ_{rs}Post_t \equiv "D_i" \times 1[t > t^*]$$

- They have a D-in-D design. They are concerned that v_{rst} may be correlated with treatment = high min. wage ($NJ_{rs}Post_t$), but assume that

$$v_{rst} = \alpha_{rs} + \epsilon_{rst}$$

so the correlation between ν_{rst} and the treatment $NJ_{rs}Post_t$ can be “differenced out” using the pre-period.

Diff-in-Diff design in Card and Krueger

Assumed the causal model is:

$$y_{rst} = \delta_1 + Post_t \delta_2^0 + NJ_{rs} Post_t \beta_1 + \alpha_{rs} + \epsilon_{rst}$$

Estimating equation:

$$y_{rst} = \delta_1 + Post_t \delta_2^0 + NJ_{rs} Post_t \beta_1 + NJ_{rs} \pi + v_{rst}$$

in which π represents $E[\alpha_{rs}|NJ] - E[\alpha_{rs}|PA]$.

With only 2 time periods, they include:

- constant
- dummy for “post” periods
- dummy for NJ restaurants
- dummy for NJ restaurants in the post period (indicator for treatment)

Diff-in-Diff estimates on Employment

- Does raising minimum wage hurt employment? A very controversial question to this day.

Average Full Time Equivalent Employment Per Restaurant, NJ and PA

	PA	NJ	NJ - PA
Mean FTE Employment, Before	23.33 (1.35)	20.44 (1.44)	-2.89 (1.44)
Mean FTE Employment, After	21.17 (0.94)	21.03 (0.52)	-0.14 (1.07)
Change in Mean FTE Emp.	-2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

- Economics could explain why employment actually goes up when min wage is higher. Labor supply curve is upward sloping (increasing in wages). Firms that set a higher wage could hire more workers!

Heterogeneity by Starting Wage

- Minimum wage increased from 4.25 to 5.05 in NJ. The increase in employment is driven by **marginal** workers who benefit from this increase.
 - the drop in employment at NJ stores unaffected by the rise in min. wage (starting wage >\$5) could be explained by the continued worsening of the economies in the middle-Atlantic states in 1992. (similar drop among higher-wage stores in PA)

Average Full Time Equivalent Employment Per Restaurant, NJ Stores with Different Wages

	Starting Wage Before Rise in Min.			Low Wage - High Wage
	\$4.25	\$4.26-4.99	>\$5.00	
Mean FTE Employment, Before	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	-2.69 (1.37)
Mean FTE Employment, After	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.67 (1.44)
Change in Mean FTE Emp.	1.32 (0.95)	0.87 (0.84)	-2.04 (1.14)	3.36 (1.48)

Diff-in-Diff with Covariates

We can generalize DD by allowing some time-varying factors z_{it} :

$$y_{it} = \delta_t + z'_{it}\beta + \alpha_i + \epsilon_{it} \quad D_i = 0 \text{ (comparison)}$$

$$y_{it} = \delta_t + z'_{it}\beta + Post_t\theta + \alpha_i + \epsilon_{it} \quad D_i = 1 \text{ (program)}$$

Following steps above we get:

$$y_{it} = \delta_t + z'_{it}\beta + D_i\pi + (Post_t \times D_i)\theta + \epsilon_{it}$$

Diff-in-Diff with Covariates

For example, Card and Krueger add controls to the diff-in-diff regression: dummies for restaurant chains, region fixed effects... The estimating equation for model 1-2 this table is

$$\Delta y_{rs} = a + b x_{rs} + c NJ_{rs} + \xi_{rs},$$

in which c on NJ represents the diff-in-diff conditional on covariates.

TABLE 4—REDUCED-FORM MODELS FOR CHANGE IN EMPLOYMENT

Independent variable	Model				
	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	2.33 (1.19)	2.30 (1.20)	—	—	—
2. Initial wage gap ^a	—	—	15.65 (6.08)	14.92 (6.21)	11.91 (7.39)
3. Controls for chain and ownership ^b	no	yes	no	yes	yes
4. Controls for region ^c	no	no	no	no	yes
5. Standard error of regression	8.79	8.78	8.76	8.76	8.75
6. Probability value for controls ^d	—	0.34	—	0.44	0.40

Notes: Standard errors are given in parentheses. The sample consists of 357 stores with available data on employment and starting wages in waves 1 and 2. The dependent variable in all models is change in FTE employment. The mean and standard deviation of the dependent variable are -0.237 and 8.825 , respectively. All models include an unrestricted constant (not reported).

^aProportional increase in starting wage necessary to raise starting wage to new minimum rate. For stores in Pennsylvania the wage gap is 0.

In models 3-5, the NJ dummy is replaced by a measure of the intensity of treatment: $(5.05 - \text{initial wage})/\text{initial wage}$, which is more positive the lower the initial wages at a restaurant before the min-wage change.

Key assumption in Diff-in-Diff

- The key assumption for DD is that the two groups would exhibit “parallel trends” in the absence of the program effect.
- With 2 or more periods in the pre- or post-periods we can evaluate the truth of this assumption.
- DD allows $E[D_i\alpha_i] \neq 0$ but *assumes* $E[D_i\epsilon_{it}] = 0$. “Selection into treatment” – i.e., $P(D_i = 1)$ – cannot depend on the transitory error components.

This rules out things like:

- management shakeup if a firm is performing worse than average
- fire coach when team has losing streak
- assign kids to different classes based on grades in previous year

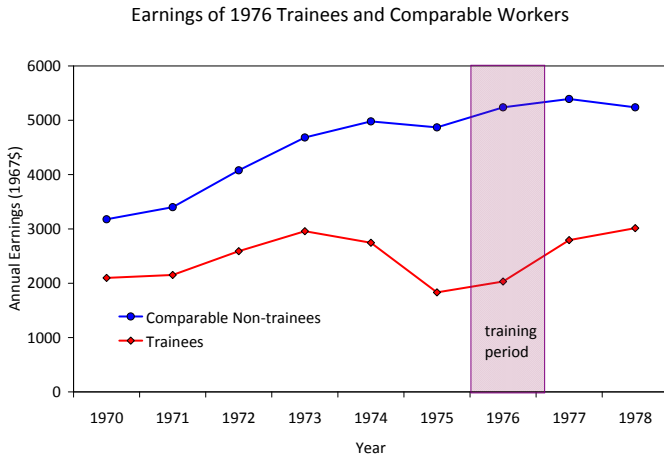
Key assumption in Diff-in-Diff

Classic violation of parallel trends:

- enrollment in retraining if a person loses a job
- often see that treatment is correlated with a transitory “dip”
- “Ashenfelter dip” (Orley Ashenfelter, 1978)

Ashenfelter Dip

- Ashenfelter and Card (1985) evaluates the effects of training programs on earnings.



Diff-in-Diff Staggered Adoption

$$y_{it} = \delta_t + z'_{it}\beta + D_i\pi + (Post_t \times D_i)\theta + \epsilon_{it}$$

We can also generalize by allowing $Post_{it}$ – the “post” dummy turns on at different times for each unit. This can be written:

$$y_{it} = \delta_t + z'_{it}\beta + D_i\pi + (Post_{it} \times D_i)\theta + \phi_i + \epsilon_{it}$$

- Staggered adoption design

Usually, what people do is estimate a slightly different model with a separate “treatment effect” for each period before/after $Post_{it}$. This is called an “event study”. Let τ_i be the period at which i is treated. So when $t = \tau_i + k$ we are k periods after i first got treated. Model is:

$$y_{it} = \delta_t + z'_{it}\beta + \sum_{k=-J}^{k=+L} \theta_k 1[t = \tau_i + k] + \phi_i + \epsilon_{it}$$

We expect θ_k for $-J \leq k \leq -1$ to be constant. This is the “pre-treatment” gap between the treatment group and the comparisons. We expect θ_k for $0 \leq k \leq L$ to be constant too. This is the “post-treatment” gap between the treatment group and the comparisons.

Example: 138 State-level Minimum Wage Changes

- Cengiz, Dube, Lindner, Zipperer (Quarterly Journal of Economics 2019) use 138 prominent min-wage changes between 1979 and 2016 in the US to estimate the effect of min-wage on low-wage jobs (employment and wages).
- To qualify for an event,
 - the state min wage was increased by $\geq \$0.25$
 - at least 2% of the workers were directly affected by the increase.
- They use a staggered diff-in-diff approach, stacking the 138 events around each's event year τ_s = the year in which there is a nontrivial min wage increase in state s . They also allow the effects to vary by every \$0.25 wage bin (*relative to the minimum wage), denoted by j below.

$$y_{sjt} = \sum_{k=-3}^{k=4} \theta_{jk} 1[t = \tau_s + k] + \delta_{st} + \mu_{sj} + \text{oth controls} + \epsilon_{sjt}$$

y_{sjt} = employment in wage bin j in state s in year t divided by the size of the population in (s, t)

Example: 138 State-level Minimum Wage Changes

- Event study estimates $\{\hat{\theta}_{jk}\}$ on average for wage bins (j)'s between [new min wage - 4, new min wage], and wage bins between [new MW, new MW + 5]. Red: the effects of a min-wage increase on the (missing) jobs below the new minimum wage. Blue: ... on the (excess) jobs at/slightly above the new min wage.

Example: 138 State-level Minimum Wage Changes

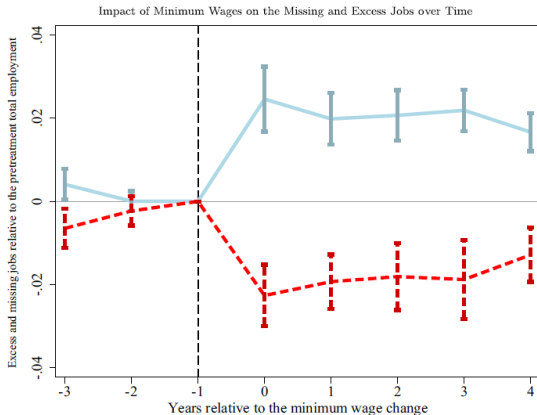


FIGURE III

Impact of Minimum Wages on the Missing and Excess Jobs over Time

Example: 138 State-level Minimum Wage Changes

- Heterogeneity: show the average $\bar{\theta}_j = \frac{1}{5} \sum_{k=1}^5 \hat{\theta}_{jk}$ effects on employment in the next 5 years by wage bins (rel. to new min wage):

EFFECT OF MINIMUM WAGES ON LOW-WAGE JOBS 1423

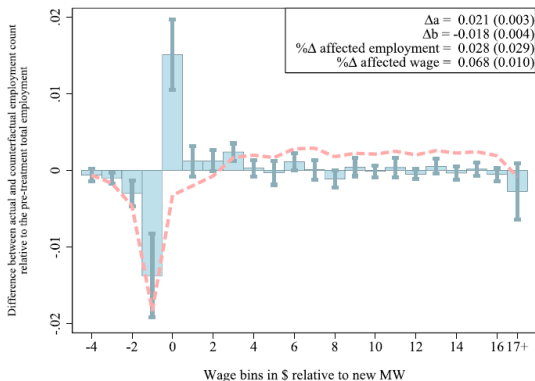


FIGURE II

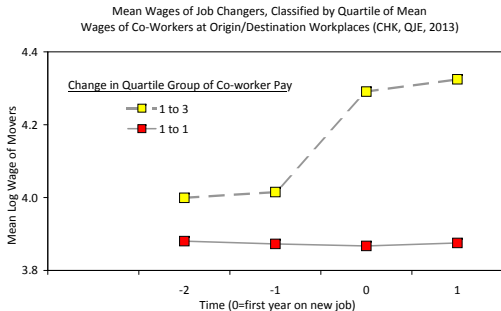
Impact of Minimum Wages on the Wage Distribution

Example: Wage Changes of Job Movers

- A simple example. Card-Heining-Kline (2013) look at German workers who change employers. So τ_i is the 1st year on the new job. They classify workers by the average pay of their co-workers in the old and new jobs.
- Firms are ranked into quartiles based on mean log wage of movers, leaving out each person's own wage.

Example: Wage Changes of Job Movers

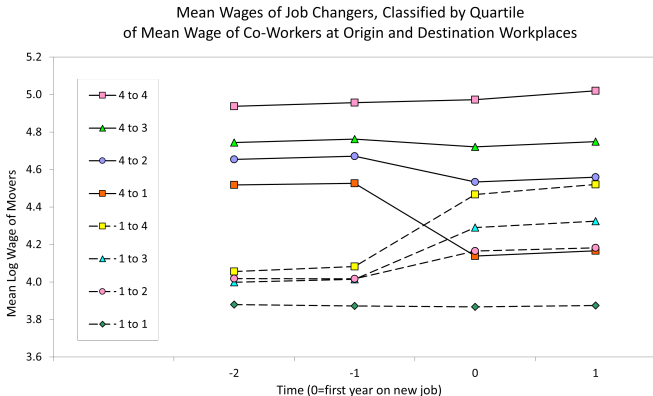
- Movers from quartile 1 firms to quartile 3 firms experience a 28 log point increase wages (compare the mean at -1 vs. 0) . No pretrend as it looks flat before the move.



Notes: figure shows mean wages of male workers observed in 2002-2009 who change jobs in 2004-2007 and held the preceding job for 2 or more years, and the new job for 2 or more years. Jobs are classified into quartiles based on mean wage of co-workers.

Example: Wage Changes of Job Movers

- The same can be shown by movers from quartile 1 (lowest wages on average) vs. movers from quartile 4 (highest wages on average):



Notes: figure shows mean wages of male workers observed in 2002-2009 who change jobs in 2004-2007 and held the preceding job for 2 or more years, and the new job for 2 or more years. "Job" refers to establishment with most earnings in year, excluding part time work. Each job is classified into quartiles based on mean wage of co-workers (quartiles are based on all full time workers in the same year).